
ME 210P: Fluid Mechanics Lab

Course Code	ME 210P
Course Name	Fluid Mechanics Lab
L-T-P-C	0-2-0-1
Prerequisites	ME 210 Fluid Mechanics
Intended for	UG
Mutual Exclusion	None
Approval	50th BoA (Jul 2023), in the B.Tech Mechanical Engineering curriculum; 53rd BoA (Dec 2023)

Course Contents

- NA

Laboratory Modules

1. Flow Visualization
2. Validation of Bernoulli's Theorem
3. Application of Flow Measuring Devices
4. Major & Minor Losses in Pipes
5. Measurement of Pipe Friction Factor
6. Identifying Losses in Pipe Fittings
7. Static Pressure Measurement in a Wind Tunnel
8. Performance Analysis of Francis & Pelton Turbine
9. Determination of Metacentric Height
10. Measurement of Lift & Drag on an Aero-foil
11. Calibration of Various Notches
12. Momentum Eqn. Verification Using Jet Impaction
13. Vortex Flow Measurement
14. Pitot Static Tube Calibration

Text Books

1. J. P. Holman, **Experimental Methods for Engineers**, 7th edition, Tata McGraw-Hill 2001.
2. T.G. Beckwith, J.H. Lienhard V, R. D. Marngoni, **Mechanical Measurements**, 5th edition, Pearson Education, 2010.
3. E.O. Doebelin, **Measurement systems, Application and Design**, 5th edition, Tata McGraw-Hill, 2008
4. Fox and Mc Donald, **Introduction to Fluid Mechanics**, 7th Edition, John Wiley, 2009

References

1. NA